RAJALAKSHMI ENGINEERING COLLEGE

Mithilesh M R 251001146-2025-IT

M2

Started on	Saturday, 5 September 2026, 12:54 PM
State	Finished
Completed on	Saturday, 5 September 2026, 12:54 PM
Time taken	32 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
int f(int a[], int l,int h,int x,int v){
    if(l>h)  return v;
    int m=l+(h-l)/2;
    if(a[m] ==x) return a[m];
    if(a[m]<x) return f(a,m+1,h,x,a[m]);
    return f(a,l,m-1,x,v);
}
int main(){
    int n;
    if(scanf("%d",&n) !=1) return 0;
    int a[n];
    for(int i=0;i<n;i++) scanf("%d",&a[i]);
    int x;
    scanf("%d",&x);
    printf("%d\n",f(a,0,n-1,x,-1));
    return 0;

}
```

	Input	Expected	Got	
✔	6 1 2 8 10 12 19 5	2	2	✔
✔	5 10 22 85 108 129 100	85	85	✔

	Input	Expected	Got	
✔	7 3 5 7 9 11 13 15 10	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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